# Supplement: Candidate-Bound Evidence for RTL-to-GDS Signoff

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## 1 Source and Execution Ledger

The primary manuscript uses frozen implementation `af31b39f6782` for fourteen design runs, twenty separate recorded-proposal executions, eighteen fault-consistency executions, and a replayed spacing-repair episode. The latter submits the recorded padding delta through a scripted proposal adapter and makes zero new model calls. It encounters native spacing failure, authorizes the recorded change, reruns the affected stages, and certifies. It supplies the post-repair artifact used by the new swap fixture. The 24 scripted baseline candidates use the same frozen native executor plus the hashed first-failure stopping harness described in the sweep protocol below. Both documents use Springer LNCS. The archive includes a portable hash verifier; native reruns require the pinned stack. Review bindings and publication receipts record separate checks.

The twenty recorded proposals originate from live Claude Sonnet 5 runs (Anthropic; identifier `claude-sonnet-5`) on snapshot `18d948815ac6`. The primary replay counts those proposal states, with six frozen certifications and fourteen native failures. It is separate from the original nine live episodes, which produced seven certifications and two escalations on that earlier code. Source snapshots, complete typed configurations, and commands identify both phases. Current source is `af31b39f6782`; Git commits modifying the manuscript and drivers do not introduce a third implementation snapshot.

The 2x2x2 padding ablation retains four source interventions derived from the frozen source. Their changes remove only the CTS and/or routing padding calls. Worker count is the other declared factor. The mapped netlist and placed DEF match across the CTS intervention, while CTS and downstream artifacts differ. The main frozen renderer retains both padding calls. Table 1 gives all eight outcomes.

## 2 Recorded Live-Repair Case Study

Three episodes per fixture type were declared before execution. Spacing and PDN each certify in 3/3 episodes; timing certifies in 1/3. The original twenty model responses record 174306 input tokens and 50164 output tokens. Table 2 retains every episode. These observations belong to `18d948815ac6`. Live repair was not re-executed on the frozen source. These are descriptive case-study outcomes for

**Table 1.** Controlled padding and thread ablation at unchanged 3 ns. Each row contains one run at each worker count. Padding refers to reapplication after placement; placement padding itself stays at one site. Slack is measured after routing.

| CTS padding | Route padding | Workers | Setup (ps) | Certified |
|---|---|---|---|---|
| Off | Off | 1, 8 | +1.586 | 2/2 |
| Off | On | 1, 8 | +9.868 | 2/2 |
| On | Off | 1, 8 | -0.654 | 0/2 |
| On | On | 1, 8 | -0.654 | 0/2 |

**Table 2.** All nine original live repair episodes on source 18d948815ac6. These archived outcomes are separate from frozen replay and larger episodes.

| Episode | Initial failure | Final evidence | Calls | Outcome |
|---|---|---|---|---|
| Spacing 1 | 1 marker | Full signoff | 1 | Verified |
| Spacing 2 | 1 marker | Full signoff | 1 | Verified |
| Spacing 3 | 1 marker | Full signoff | 1 | Verified |
| PDN 1 | PDN-0185 | Full signoff | 1 | Verified |
| PDN 2 | PDN-0185 | Full signoff | 1 | Verified |
| PDN 3 | PDN-0185 | Full signoff | 1 | Verified |
| Setup 1 | -0.326192 ns | 0.001586 ns | 2 | Verified |
| Setup 2 | -0.326192 ns | Not certified | 3 | Escalated |
| Setup 3 | -0.326192 ns | Not certified | 9 | Escalated |

repeated fixtures, with no population repair-rate estimate or claim of frozen-source live-repair success.

The original timing sequence at unchanged 3 ns is shown in Figure 1. The successful configuration repeats three times at +1.586 ps under its archived source and three times at −0.654 ps under the frozen implementation. The current-source failure is retained; the primary factorial experiment measures the effect of CTS padding rather than attributing the result to uncontrolled noise.

The archived class replay gives spacing 0/3 failed states, PDN 0/3, and timing 11/14, with two additional timing states lacking a completed target check. Every confirmed failure in the earlier 11/20 headline is therefore from the timing fixture. Fresh native execution establishes outcomes for all twenty states on the frozen source: spacing 0/3 failures, PDN 0/3, and timing 14/14. No unavailable target is silently counted as a confirmed archived failure. Excluding `timing_1_proposal2`, the CTS-affected archived pass, leaves twelve setup STA failures and one placement failure among thirteen frozen timing states. The placement-failed state never reaches routed STA.

## 3 Complete-State Replay Reconstruction

Reconstruction starts with the archived IR defaults and applies the logged characterization seed, initial overrides, curated numerical changes, and all accepted

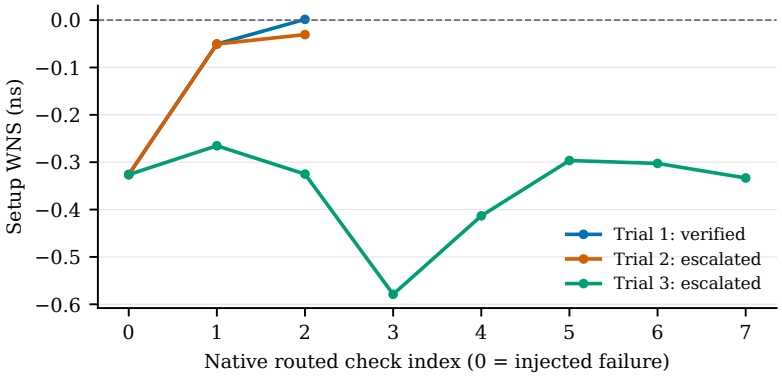



**Fig. 1.** Archived native routed timing checks at 3 ns on source 18d948815ac6. Lines end at the last completed check, including escalated episodes.

model deltas in temporal order. The archived renderer used the native one-worker default; the current replay declares that effective count explicitly. All twenty reconstructed inputs are consistent with archived defaults and logged changes. Independently retained full IR records exactly match seven final proposal states (three spacing, three PDN, one timing) before adding `routing.num_threads=1`, which was absent from the archived schema. No independent complete IR record for the other thirteen states was found in the inspected archive; attempt snapshots retain individual sections. Thus 20/20 trace consistency and 7/20 independent full-state comparisons are different checks. The per-state inventory is retained in `rtl2gdsagi_paper/evidence/replay_stratified_audit.json`. Fresh runs disable further repair.

An initial reconstruction pilot omitted the curated PDN changes preceding the model call: strap offset 50 to 21.42 $\mu$m and utilization 0.35 to 0.29. Its three PDN configurations did not represent the recorded promoted states. The complete pilot and original launcher remain in the archive and are excluded from the final twenty-state result. The corrected cohort was declared and executed separately, with the full-state equality checks described above. This exclusion corrects configuration reconstruction rather than selecting favorable native outcomes.

## 4 Scripted Sweep and Stopping Policy

The 24-state sweep derives initial configurations before the first failed native check of each recorded fixture. It applies fixed numerical choices, never model responses: padding 0–3, offset 50/2 $\mu$m, and the full sixteen-state DELAY0/DELAY1, density 0.45/0.55, padding 1/2, setup-repair off/on grid at 3 ns. CTS optimization margins stay at 0.05 ns; the acceptance guardband stays zero. Two period controls use the initial AREA0 timing state at 3.25 and 3.50 ns. They keep delay fractions 0.2, which changes absolute input/output delay

from 0.60 ns to 0.65/0.70 ns; uncertainty and load remain fixed. These are combined requirement changes, excluded from fixed-target repair counts. All eight setup off/on pairs yield identical CTS/routed DEF, SPEF and setup/hold WNS. Diagnostic re-execution of the eight setup-on CTS scripts retains their inputs and command order, adding only slack markers and redirected output snapshots. Every setup call is entered and returns normally. Setup WNS immediately before and after the call is unchanged at 194.356–274.527 ps, above the 50 ps target. The before/after DEF and netlist hashes match, and every final diagnostic CTS DEF matches its original candidate. The scripts set clock-layer RC but omit signal-wire RC; native warnings RSZ-0014/RSZ-0021 report zero wire capacitance and wire-load fallback. Thus no setup violation is visible at the repair point under that estimator, while later routed SPEF timing fails. This limits what the toggle experiment establishes about setup repair. The eight logs, scripts, markers and snapshot hashes are retained in `runs/setup_repair_diagnostic_20261013`; these are diagnostics, not additional end-to-end candidates or certifications.

The original no-model CLI pilot repeatedly rerouted the unpadded candidate after DRC rejection: date-bearing GDS changed the failing-input fingerprint without an IR change. It was stopped and retained, with its completed runs excluded. A bootstrap import failure created no native results and is also retained. The final cohort was declared separately. Its study-only `scripts/run_baseline_candidate.py` subclasses the frozen controller solely to stop when exhaustion would trigger rollback. Every native gate, parser and successful certification path remains frozen. Known PDN failure also terminates under the manual policy. This is a bounded candidate-acceptance experiment, not an iterative model-versus-controller repair comparison.

`rtl2gdsagi_paper/evidence/scripted_baseline_metrics.csv` reports every candidate, full certification, first failed stage, measured setup/hold slack, DRC markers and elapsed time. A metric is "not reached" when the native check was not executed. Configuration and harness hashes are retained in `runs/scripted_baseline_bounded_20261013/experiment_manifest.json`. Successful candidate/certification identities are in each run's `signoff.json` and `gate_evidence.json`. Failed states retain configuration and native input records, without release certificates.

## 5 Guardband Sensitivity from Retained Slacks

The acceptance guardband in every recorded run is $g = 0$. We calculate eligibility at $g = 0.05$ ns without modifying implementation, constraints, reports or certificates. Post-CTS hold and routed setup/hold must all reach that threshold; every other obligation retains its recorded outcome. Table 3 reports the seven primary designs and the archived narrow timing repair. All fourteen primary runs pass at $g = 0$; twelve (six designs) remain eligible at 50 ps. Both shift1024 repeats fail routed hold, with 49.477 ps against 50 ps; post-CTS hold still passes. The archived +1.586 ps setup repair fails the stricter setup threshold.

**Table 3.** Post-hoc 50 ps guardband sensitivity. All listed candidates were certified at $g = 0$; both primary repeats have these same slacks. Eligibility also requires post-CTS hold. No new certificates are issued.

| Candidate | Post-CTS hold (ps) | Routed setup (ps) | Routed hold (ps) | $g = 50$ ps |
|---|---|---|---|---|
| register | 1220.036 | 6954.046 | 1203.611 | Pass |
| mux4to1 | 1220.036 | 6951.572 | 1202.738 | Pass |
| alu8 | 1214.006 | 5256.096 | 1191.852 | Pass |
| counter | 476.518 | 4910.823 | 493.969 | Pass |
| shift_reg | 528.552 | 6891.726 | 553.843 | Pass |
| fifo_buf | 259.698 | 6395.702 | 272.794 | Pass |
| shift1024 | 65.326 | 5553.176 | 49.477 | Fail |
| Archived timing repair | 322.043 | 1.586 | 343.180 | Fail |

Among already certified states in the other separate cohorts, 6/7 archived live certifications, 6/6 frozen replay certifications, 5/5 scripted baseline certifications, and 0/4 padding-ablation certifications remain eligible. These cohort-specific counts are not pooled: they reuse configurations and have different execution/proposal origins. The archived live analysis omits two noncertified episodes without terminal certified-candidate timing. `rtl2gdsagi_paper/evidence/guardband_sensitivity_metrics.csv` retains all 73 inspected states, checkpoint availability, original certification, and threshold outcomes. Missing STA remains "not reached"; a timing pass cannot certify a state rejected by another gate. This post-hoc sensitivity does not measure robustness across corners, process variation or native reruns.

## 6 Native Checker Invocation Controls

The pinned OpenLane wrapper and public source bytes match. The installed modern PDK deck expects runtime rule flags that this wrapper does not forward. The previous pairing declares zero categories and zero markers for both known violating and clean synthetic geometry. It is an inactive configuration observation and is excluded as an active-DRC baseline.

Table 4 reports active paired controls. The legacy native deck reports 166 categories and metal/off-grid violations; its tested poly category is absent. The modern PDK options and explicit adapter each report 220 categories, with metal/poly/off-grid markers 3/1/4 and paired clean counts 0/0/0. These control counts are raw inventories for those configurations, not the primary controller's 211 normalized-category contract. Violating stage calls return zero despite markers, so marker reporting and native exit rejection are distinguished. The adapter is a recorded sidecar change, not an unmodified stock full-flow run.

The selected PDK options enable FEOL, BEOL, off-grid and seal checks and disable floating metal. The direct modern command passes `feol=1`, `beol=1`, `offgrid=1`, `seal=1`, and `floating_met=0`. Exact argument vectors are retained

**Table 4.** Paired synthetic DRC controls. Marker columns list metal/poly/off-grid fixtures. Modern controls use the PDK runtime options; the adapter is an explicit sidecar change. The inactive prior wrapper is a configuration diagnostic, excluded from this active-checker table.

| Executed configuration | Categories | Violating markers | Clean markers |
|---|---|---|---|
| Legacy native deck | 166 | 3/0/4 | 0/0/0 |
| Modern deck, PDK options | 220 | 3/1/4 | 0/0/0 |
| Modern wrapper adapter | 220 | 3/1/4 | 0/0/0 |

in `runs/openlane_invocation_audit/controls_v1/executions/*/command.json`; `rtl2gdsagi_paper/evidence/openlane_invocation_audit.json` binds their inputs and reports. The retained formal controls use EQY revision `v0.68-2-g4a72eb9`. Comprehensive rule coverage, default Magic checking, and a full OpenLane-flow comparison remain unassessed. Native supply checks from the separate retained study detect disconnected vias with paired untouched DEFs. Optional reference XOR detects geometry differences when supplied a reference. The retained KLayout no-compare database is incompatible with the Netgen report interface and is not a semantic detection control. The stock source and PDK configuration provide public invocation evidence; no personal maintainer endorsement is asserted.

## 7 Exploratory Larger Episodes and Stopping Proposals

The 1,995-cell `accum_bank32` contains thirty-two eight-bit accumulator lanes and a read-selection multiplexer. Two spacing and two antenna episodes were declared on `af31b39f6782`. Table 5 retains all four: 0/4 certify, two are policy escalations on previously failed fingerprints, and two are credit-interrupted and remain incomplete. The latter outcomes stay unknown; their saved candidate states are not certified or labeled eventual algorithmic repair failures. Twenty-three valid model responses are retained across this separate cohort. It is not pooled with the original nine episodes.

**Table 5.** All four exploratory 1,995-cell accumulator episodes on source af31b39f6782. Responses are valid recorded model returns. No episode completes certification. Incomplete episodes have unknown eventual repair outcomes and are not classified as algorithmic failures.

| Case | Responses | Last failed native stage | Recorded outcome |
|---|---|---|---|
| Antenna 1 | 3 | routing | Controller stop |
| Antenna 2 | 3 | routing | Credit-interrupted |
| Spacing 1 | 10 | placement | Credit-interrupted |
| Spacing 2 | 7 | drc | Controller stop |

The spacing-stop episode's last proposal raises padding from four to six sites. A previous six-site attempt at the same density already failed placement with DPL-0036. The sequence cycles through padding two, four and six; the recorded earlier DRC counts are one and four markers before the six-site placement failure. The controller rejects revisiting that failed placement fingerprint after 7/24 diagnosis attempts.

The antenna-stop episode first increases repair iterations to ten and the prediction margin to 40%, then changes maximum routing layer to six and the detailed-router iteration limit to 64. Layer six causes GRT-0005. The final proposal restores maximum layer five. The resulting antenna-stage inputs revisit the prior failed signature, which had three violating nets/pins, and the controller stops after 3/24 diagnosis attempts. These are revisited failed states rather than global-budget exhaustion or an unchanged literal final delta.

Earlier spacing, antenna and diode-legalization probes of the accumulator, ALU banks, and other scale candidates remain in the native archive. They are excluded from the seven clean primary designs. Exact mapped-cell counts, failed stages, and intermediate observations are preserved in the JSON catalog.

## 8 Earlier Retained Observations

Table 6 preserves earlier observations under their own protocols. They do not enlarge the prospective nine-episode denominator or the twenty-state frozen replay. Full logs record the actual terminal state; unavailable intermediate prompt/response or artifact files are not invented.

**Table 6.** Earlier retained observations. Protocols and source snapshots differ; these records remain separate from the prospective nine-episode cohort. The JSON catalog provides the exact run paths and raw observations.

| Case | First requested stage | Responses | Native certification |
|---|---|---|---|
| H1 | drc | 1 | Verified |
| H2 | drc | 0 | Not certified |
| H3 | drc | 1 | Verified |
| H4 | drc | 1 | Verified |
| H5 | sta_signoff | 0 | Not certified |
| H6 | sta_signoff | 0 | Not certified |
| H7 | sta_signoff | 1 | Not certified |
| H8 | sta_signoff | 0 | Not certified |
| H9 | lint | 0 | Not certified |
| H10 | lint | 1 | Not certified |